

LAB - 1

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DIV - CS1

BATCH - C13

Practical 1

About this unit

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1.1.1. Calculate Momentum

32:22

Write a program that accepts the mass of an object (in kilograms) and its velocity (in meters per second), then calculates and displays the momentum of the object. The momentum p is calculated using the formula:

$$p = m \times v$$

where:

m is the mass of the object (in kilograms).

v is the velocity of the object (in meters per second).

Input Format:

- A single floating-point number representing the mass of the object in kilograms.
- A single floating-point number representing the velocity of the object in meters per second.

Output Format:

- The output will display calculated momentum with appropriate units (kgm/s) (rounded up to 2 decimal places).

Sample Test Cases

+

calculate...

Submit

```
1 mass=float(input(""))
2 velocity=float(input(""))
3 momentum=mass*velocity
4 print(f"{momentum:.2f}kgm/s")
```

Terminal

Test cases

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19:59
04-05-2026

1.1.2. Conditional Calculation Based on the Number of Digits

09:04

Write a Python program that accepts an integer n as input. Depending on the number of digits in n .

Constraints:

$1 \leq n \leq 999$

Input Format:

- The input consists of a single integer n .

Output Format:

- If n is a single-digit number, print its square.
- If n is a two-digit number, print its square root (rounded to two decimal places).
- If n is a three-digit number, print its cube root (rounded to two decimal places).
- Else print "Invalid".

Sample Test Cases

+

condition...

```
1 #write your code here...
2 n = int(input(""))
3 if (n>0 and n<10):
4     square = n*n
5     print(square)
6 elif (n>=10 and n<100):
7     sqrt = n**(1/2)
8     print(f"{sqrt:.2f}")
9 elif (n>=100 and n<=999):
10    cbt = n**(1/3)
11    print(f"{cbt:.2f}")
12 else:
13    print("Invalid")
```

Terminal

Test cases

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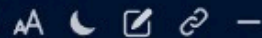


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19:59
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1.1.3. Days between Two Dates

20:02



Write a Python program to calculate number of days between two dates.

Input Format:

- The first line contains the first date in the format *YYYY-MM-DD*.
- The second line contains the second date in the format *YYYY-MM-DD*.

Output Format:

- An integer representing the number of days between the given dates.

Note:

- The first date should always be considered earlier than the second date.

Sample Test Cases



Explorer

noOfDay...



Submit

Debugger

```
1 from datetime import date
2 from datetime import datetime
3
4 #write your code here...
5 date1 = input().strip()
6 date2 = input().strip()
7
8 d1 = datetime.strptime(date1, "%Y-%m-%d")
9 d2 = datetime.strptime(date2, "%Y-%m-%d")
10
11 difference = d2 - d1
12 print(difference.days)
13
14
```

Terminal

Test cases

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20:00
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1.1.4. Reverse a Number

Write a Python program to reverse the digits of the number and print the reversed number.

Input Format:

- The input is a single integer.

Output Format:

- The output prints a single integer, which is the reversed number.

Sample Test Cases

+

reverseN...

```
1 #write your code here...
2 num=int(input()) #read the string input
3 reverse_num=str(num)[::-1]
4 print(int(reverse_num)) #return value from string to integer
```

Terminal

Test cases

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20:00

04-05-2026

1.1.5. Multiplication Table

18:13

Write a Python program that takes an integer as input and prints the multiplication table for that integer from 1 to 10.

Input Format:

- The first line of input contains an integer that represents the number for which the multiplication table is to be printed.

Output Format:

- Print the multiplication table for the given number in the format:

<number> x <multiplier> = <result>

Note:

- The character "x" is used in the output to represent multiplication.
- Refer to the visible test-cases for more clarity.

Sample Test Cases



Explorer multiplic...

```
1 #write your code here...
2 n = int(input(""))
3 for i in range(1,11):
4     print(n, "x", i, "=", n*i)
5
6
```

Debugger

Terminal Test cases

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20:00
04-05-2026



1.2.1. Pass or Fail

09:22 A M E P -

Write a Python program that accepts the number of courses and the marks of a student in those courses.

The grade is determined based on the aggregate percentage:

- If the aggregate percentage is greater than 75, the grade is Distinction.
- If the aggregate percentage is greater than or equal to 60 but less than 75, the grade is First Division.
- If the aggregate percentage is greater than or equal to 50 but less than 60, the grade is Second Division.
- If the aggregate percentage is greater than or equal to 40 but less than 50, the grade is Third Division.

Input Format:

- The first input will be an integer n , the number of courses.
- The second input will be n integers representing the marks of the student in each of the n courses, separated by a space.

Output Format:

If the student passes all courses:

- Print the aggregate percentage (formatted to two decimal places).
- Print the grade based on the aggregate percentage.

If the student fails any course (marks < 40 in any course), print:

- "Fail".

Note:

- Aggregate Percentage: Average performance across all courses, calculated by summing all marks and dividing by the number of courses

passorFa... Submit

```
1 # write your code here
2 n = int(input())
3 marks = list(map(int, input().split()))
4 if any(mar < 40 for mar in marks):
5     print("Fail")
6 else:
7     aggregate_Percentage = sum(marks) / n
8     print(f"Aggregate Percentage: {aggregate_Percentage :.2f}")
9     if aggregate_Percentage > 75:
10        print("Grade: Distinction")
11    elif 60 <= aggregate_Percentage < 75:
12        print("Grade: First Division")
13    elif 50 <= aggregate_Percentage < 60:
14        print("Grade: Second Division")
15    elif 40 <= aggregate_Percentage < 50:
16        print("Grade: Third Division")
17
```

Terminal Test cases

1.2.2. Fibonacci Series

Write a Python program that uses recursion to print the first n terms of the Fibonacci series.

Input Format:

- A single integer n representing the number of terms to generate.

Output Format:

- A single line containing the first n terms of the Fibonacci sequence, separated by spaces.

Sample Test Cases

```
1 def fibonacci(n):
2
3     if (n == 1):
4         return 0
5     elif (n == 2):
6         return 1
7     else:
8         return fibonacci(n-1) + fibonacci(n-2)
9
10
11 n = int(input())
12 for i in range(1, n + 1):
13     print(fibonacci(i), end=" ")
14
```

Terminal Test cases

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1.2.3. Pattern - 1

09:35

Write a Python program to print a pattern of asterisks in the form of a right-angled triangle.

Input Format:

- The input is an integer, representing the number of rows in the pattern.

Output Format

- The output should display the pattern of asterisks (*), with each row containing an increasing number of asterisks.

Note: Refer to the displayed test cases for the sample pattern.

Sample Test Cases

rightangl...

```
1 #Type Content here...
2 n = int(input())
3 for i in range(1,n+1):
4     for j in range(1,i+1):
5         print("*",end=" ")
6     print()
7
```

Submit

Terminal

Test cases

1.2.4. Pattern - 204:42

Write a Python program to print a right-angled triangle pattern of numbers.

Input Format:

- The input is an integer, representing the number of rows in the pattern.

Output Format:

- The output should display the pattern of numbers separated by space, with each row containing increasing numbers starting from 1 up to the row number.

Note:

- Refer to the displayed test cases for the sample pattern.

Sample Test Cases

numberP...

1 n = int(input())
2 for i in range(1, n+1):
3 for j in range(1, i+1):
4 print(j, end = " ")
5 print()
6
7

Submit